

7. Which of the following instrument measures blood pressure –

- (a) Spherometer (b) Anemometer
(c) Sphygmomanometer (d) Ammeter

48th to 52nd B.P.S.C. (Pre) 2008

Ans. (c)

A Sphygmomanometer is a device used to measure blood pressure, composed of an inflatable cuff to restrict blood flow and a mercury or mechanical manometer to measure the pressure. It is always used in conjunction with a means to determine at what pressure blood flow is just starting and at what pressure it is unimpeded. Manual sphygmomanometers are used in conjunction with a stethoscope.

8. The blood group of universal donor is –

- (a) B (b) O
(c) A (d) AB

40th B.P.S.C. (Pre) 1995

43rd B.P.S.C. (Pre) 1999

Ans. (b)

Individuals with blood group O (Negative) are a universal donor because their red blood corpuscles have neither A nor B antigens on their surface. So, the blood of a person having O (Negative) group can be given to people with any blood group.

9. The universal acceptor blood group is :

- (a) A (b) B
(c) AB (d) O
(e) None of the above / more than one of the above

67th B.P.S.C. (Pre) 2021

Ans.(c)

AB (Positive) blood group is the universal acceptor while O (Negative) is the universal donor blood group.

10. In which organ of the human body are the lymphocytes formed?

- (a) Liver (b) Bone marrow
(c) Spleen (d) Pancreas
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (e)

The functional lifetime of an erythrocyte is about 100–120 days, during which time the erythrocytes are continually moved by the blood flow push (in arteries), pull (in veins) and a combination of the two as they squeeze through microvessels such as capillaries. The spleen is an organ found in virtually all vertebrates. Similar in structure to a large lymph node, it acts primarily as a blood filter. The spleen plays important role in regard to red blood corpuscles (also referred

to as erythrocytes) and the immune system. It removes old red blood corpuscles and holds a reserve of blood, which can be valuable in case of haemorrhagic shock and also recycles iron. In the human adult, the bone marrow alone produces all of the RBCs, 60-70 percent of the WBCs and all of the platelets. The bone marrow and the lymphatic tissues, particularly the spleen, the thymus, and the lymph nodes, produce the lymphocytes (comprising 20-25 percent of WBCs).

11. In Tachycardia, the heart rate becomes

- (a) slow (b) normal
(c) fast (d) More than one of the above
(e) None of the above

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Ans. (c)

Tachycardia is the medical term for a heart rate over 100 beats a minute. Many types of irregular heart rhythms, called arrhythmias, can cause tachycardia.

12. A drug which reduces high blood pressure is obtained from

- (a) Solanum nigrum (b) Aconitum chasmanthum
(c) Centella asiatica (d) Rauwolfia serpentina

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Ans. (d)

Rauwolfia Serpentina alkaloids work by controlling nerve impulses along certain nerve pathways. As a result, they act on the heart and blood vessels to lower blood pressure.

V. Excretory System

1. The filtration unit in human kidney is called as

- (a) Nephron (b) Neutron
(c) Neuron (d) More than one of the above
(e) None of the above

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Ans. (a)

In humans, the excretory system consists of a pair of kidneys, one pair of ureters, a urinary bladder and a urethra. Each kidney has nearly one million complex tubular structures called nephrons, which are the functional units. The filtration unit in human kidney is called as Nephron.

2. The process by which blood is purified in human body is called

- (a) Dialysis (b) Haemolysis
(c) Osmosis (d) Paralysis

44th B.P.S.C. (Pre) 2000

Ans. (a)